LEARNER GUIDE

Cross-sections

2D and 3D Shapes. A short guide to walk beside you — read as much or as little as helps.
Connection before curriculum. Always.

What this lesson is about

A cross-section is the flat 2D shape you get when you slice straight through a solid. In this lesson you build a prism by carrying a 2D shape along a length, slice all kinds of solids and watch the 2D shape appear face-on, and discover that the same solid can give different shapes depending on how you cut it. You will meet a special family of solids called prisms — solids with the same cross-section all the way along — and even see a cone reveal a circle, an ellipse and a parabola. The question you are exploring is: what shape appears when you slice straight through a solid?

What you are learning to notice

- That a cross-section is the 2D shape revealed when you slice through a solid.
- That a prism is a solid with the same cross-section all along its length.
- How the direction of a cut changes the shape — a cylinder gives a circle, a rectangle or an ellipse.
- That slicing a cone can reveal a circle, an ellipse or a parabola.

Moving through it, one step at a time

In the Extruder, choose a 2D shape and carry it along a length to build a prism — notice every slice stays the same. In the Slicing Plane Studio, pick a solid and drag the height and tilt of the cut; the cross-section appears face-on beside it. In Name the Slice, predict the 2D shape from how the solid is cut. Then design your own prism and predict a slice halfway along it. There is no rush; use the isometric grid on the Scratchpad to sketch solids and their slices.

A gentle reminder
You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the names and properties of basic 2D shapes (L1) and the idea of a solid's faces), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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